REFLOWECONOMY · STEM CHECK

ReflowEconomy STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/refloweconomy		
Source	docs/01b-stem-check.md @ 9819cf9		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First STEM check at the kit 1.9.1 upgrade (repo already at TRL 3); S13 and S17 fixed under Amish's standing instruction of 2026-10-23 (RFE-DEC-001 D2, D3); S14 to Amish (D1); independent review recorded

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Verdict: the reference micro-factory (option A) sits within known physics, chemistry, biology and mathematics on its mass balance, float-sink separation, press heating, fume capture, duct and fan, guarding, structure and economics, and nothing fails outright, but the check is **open**. One claim fails in detail and its fix needs Amish: counted as currents rather than kilowatts, the morning block of loads draws about 55 A on paper from a 230 V, 40 A single-phase supply (S14). Two detail failures were fixed in this session under Amish's standing instruction of 2026-10-23: motor energy is now counted at the electrical input, which moves the line from 0.83 to 0.88 kWh per kilogram of output (S13), and the feed slot is lowered from 160 to 120 mm so the ISO 13857 reach distance applies to it (S17). Five claims stay plausible rather than proven after the independent review: the press standing loss (S3), noise (S15), the feed reach until the machine-specific EN 12012 standard is checked (S17), manual lifting under real conditions (S20) and air quality (S21); most need measurements on a running site, which is TRL 4 work and on hold. Option B, the sort-and-bale hub, sits within known science; its bale density is an assumption to confirm (B1). Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the calculation note's script `docs/04-calcs/sizing.py` (RFE-CAL-001) so every figure reruns with the design.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` by `docs/04-calcs/stem_check.py`).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Every kilogram of the 100 kg input leaves as products (32.3), flake (19.3), metals, e-waste, paper or disposal (28.4)	Mathematics	Numerical method	Conservation of mass (RFE-CAL-001 section 2)	in 100 kg, out 100 kg	Within known science	RFE-CAL-001 section 2
S2	In water, PP and HDPE flake floats and PET and PVC sink, so float-sink separates them (it cannot split HDPE from PP, which are sorted into their own bins at the table)	Physics	Fluid mechanics	Archimedes: a flake floats if its density is below water's (1.00 g/cm3); HDPE 0.94 to 0.97, PP 0.90 to 0.91, PET 1.33 to 1.45, PVC 1.38 to 1.41 g/cm3 (KiTech; recipes)	0.97 g/cm3 <= 1 g/cm3	Within known science	docs/ playbook/ recipes/pet-flake.md
S3	The hot sheet press loses about 0.8 kW standing when hot	Physics	Heat transfer	Conduction through 50 mm mineral wool (k 0.045 W/m K) over the platen packs at 175 K, plus 50 % for edges and frame; closed press only (losses while loading and through the frame and jacks not counted)	claimed 0.8 kW, estimate 0.717 kW (ratio 1.12)	Plausible, to confirm at TRL 3	Open: measure the standing loss on the built press (TRL 4, on hold)
S4	At full heater power the 5 kW press puts a 12 mm HDPE sheet's heat in within the published 55 min cycle (cooling is in the separate cooling press)	Physics	Thermodynamics	Energy balance: sheet and mould heat 2.56 kWh (HDPE cp 2.25 kJ/kg K over 175 K plus 180 kJ/kg fusion, about 62 % of the 293 J/g of fully crystalline PE) at 5.0 kW less the 0.8 kW standing loss;	36.6 min <= 55 min	Within known science	RFE-CAL-001 section 4

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				average draw over the cycle 3.6 kW			
S5	Heat reaches the middle of a 12 mm sheet well inside the 55 min cycle	Physics	Heat transfer	Transient conduction t = L^2 / alpha (alpha 0.21 mm2/s for HDPE), x1.5 for the latent heat, x4 for a loose flake bed	17.1 min <= 55 min	Within known science	RFE-CAL-001 section 4
S6	HDPE and PP are processed at 190 to 210 C, below the 230 C (HDPE) and 240 C (PP) ceilings of their processing ranges	Chemistry	Materials and polymer chemistry	Processing ranges HDPE 190 to 230 C, PP 200 to 240 C (RJC Mold chart; Precious Plastic Extrusion Pro)	210 C <= 230 C	Within known science	docs/ playbook/ recipes/ hdpe.md, pp.md
S7	PVC is never heated: it releases hydrogen chloride at press temperatures, and float-sink cannot remove it from PET	Chemistry	Materials and polymer chemistry	PVC dehydrochlorination: 0.4 mol % HCl lost in 30 min at 200 C (Szarka and Ivan, 2013); PVC 1.38 to 1.41 g/cm3 sinks with PET	Exclusion rule at sorting (safety rules, recipes, BLD stop S6); within known chemistry	Within known science	docs/ playbook/ safety.md
S8	PET is sold as flake, not melted: it must be dried to about 0.005 % moisture at 150 to 175 C for 4 to 5 h or it hydrolyses in the melt	Chemistry	Materials and polymer chemistry	Hydrolytic degradation of PET in the melt (Plastics Technology, How to Dry PET)	Design choice decided by Amish, 2026-09-25 (RFE-DDR-001 item 3); consistent with the chemistry	Within known science	RFE-DDR-001
S9	Polyolefin melt fumes contain formaldehyde, acetaldehyde, acrolein, CO and a wax	Chemistry	Environmental and atmospheric chemistry	Borealis, Statement on impurities: polyolefins, ed. 6, 2022	Two enclosing hoods, G4 and F7 filters, impregnated carbon, stack	Within known science	RFE-CAL-001 section 6

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	aerosol, so every melt process needs local exhaust				above the roof (RFE-CAL-001 section 6)		
S10	The press booth's extract (0.5 m/s at the sash) carries away the hot-platen plume, so fume does not spill out of the sash	Physics	Fluid mechanics	Axisymmetric plume mass flow m = 0.071 Qc^(1/3) z^(5/3) + 0.0018 Qc (NFPA 92, a fire-plume equation used here as a heuristic well below its range), Qc from natural convection off the open 1.1 m platen, z to the booth roof; a hot mould being unloaded is a larger, short source	0.132 kg/s <= 0.475 kg/s	Within known science	RFE-CAL-001 section 6
S11	The 250 mm duct runs at a transport velocity that keeps the wax aerosol from settling	Engineering and technology	Standards practice	Duct design velocity: ACGIH gives about 5 to 10 m/s for vapours and fumes; 10 m/s is used here so the condensed wax aerosol does not settle (design choice, both hoods drawing)	11.7 m/s >= 10 m/s	Within known science	RFE-CAL-001 section 6
S12	The 1.5 kW fan moves 0.57 m3/s against 815 Pa with dirty filters	Engineering and technology	Power and link budgets	Fan input = Q dp / (fan 0.55 x motor 0.85)	1 kW <= 1.5 kW	Within known science	RFE-CAL-001 section 6
S13	The line uses 1.0 kWh or less per kilogram of output with motor and drive losses counted (was 0.83 on shaft power)	Engineering and technology	Power and link budgets	Electrical energy = shaft energy / (motor x drive); IE3 minimum efficiencies, IEC 60034-30-1 (ABB note): 2.2 kW 86.7 %, 1.5 kW 85.3 %; drive 96 %; single-phase pump 75 %	0.88 kWh/kg <= 1 kWh/kg	Within known science	RFE-DEC-001 D2 (fix made 2026-10-25: CAL counts motor and drive losses)
S14	The block A loads run on a 230 V, 40 A	Physics	Electromagnetism and circuits	I = P / (V x PF x eta); drive input currents from datasheets	Block A 55.4 A running (72.0 A at	Fails, fix proposed	Open: RFE-DEC-001 D1, proposed,

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	single-phase supply (the old figure was 8.75 kW = 38.0 A at unity power factor)			(Schneider ATV12: 20.2 A at 2.2 kW, 14.9 A at 1.5 kW, 240 V); single-phase motor currents NEC Table 430.248	full load; arithmetic sum of table currents, a conservative upper estimate) against 40 A; block B 36.4 A, 91 % of the supply as a continuous load. Fixes: three-phase 400 V supply (18.5 A per phase if balanced) or three time blocks on single phase (largest 36.4 A, about 10.6 h day, about 13 USD a shift more in wages)		awaiting Amish
S15	With the lined enclosure the shredder operator's 8 h exposure is 85 dB(A) or less (sound power assumed 102 dB(A))	Physics	Acoustics and vibration	Lp = Lw + 10 log(Q / 4 pi r^2 + 4 / R) in the reverberant steel room; 15 dB insertion loss; LEX over 3.79 h	79.1 dB(A) <= 85 dB(A)	Plausible, to confirm at TRL 3	Open: measure the machine's sound power and the enclosure on site (TRL 4, on hold)
S16	A 15 dB insertion loss is possible with the feed slot open	Physics	Acoustics and vibration	Open-area limit on insertion loss IL <= 10 log(S_enclosure / S_open)	15 dB <= 24.1 dB	Within known science	RFE-CAL-001 section 7
S17	A hand through the feed slot cannot reach the cutters: the slot is 120	Engineering and technology	Standards practice	ISO 13857 Table 4: slot 40 < e <= 120 mm, safety distance 850 mm; above 120 mm the reaching-over distances apply	1060 mm >= 850 mm	Plausible, to confirm at TRL 3	Open: fix D3 made 2026-10-25 (slot 160 to 120 mm); the EN

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	mm high (ISO 13857 Table 4 range) and the path is 1,060 mm against 850 mm			(Troax and Axelent guides to ISO 13857)			12012 type C standard for size reduction machines, which takes precedence over ISO 13857, still to check; cutter height assumed
S18	A UB 305 x 165 x 40 that fits the reserved 165 x 305 mm space carries the roof edges over the 12 m seam (top flange held by the roof)	Physics	Strength of materials	Bending only: M = w L^2 / 8 at ULS (1.35 g + 1.5 q; roof 0.20 kN/m2 plus 0.40 kN/m beam, imposed 0.75 kN/m2 over 2.44 m) against Wpl fy (623 cm3, S355). Unrestrained it would buckle laterally, so the top flange must be tied to the roof; service deflection about 41 mm (L/290) misses L/360, and racking stiffness lost with the side walls is not checked: both for the structural engineer	71.3 kN m <= 221.2 kN m	Within known science	Structural engineer's design (RFE-DEC-001 confirm item 1)
S19	The container floor carries the 430 kg hot press	Physics	Mechanics and statics	Line load on a 40 ft container floor, 3.0 t/m allowed (TIS, ISO 1496 limits); press over 1.2 m	0.36 t/m <= 3 t/m	Within known science	RFE-CAL-001 section 8
S20	No one lifts more than 25 kg: flake and product lots are packed in 25 kg bags or moved on the transfer bridge	Biology	Ergonomics	ISO 11228-1 reference mass 25 kg is for ideal conditions; frequency, height (the feed lip is at 1.70 m) and reach multipliers reduce it	25 kg <= 25 kg	Plausible, to confirm at TRL 3	Open: apply the ISO 11228-1 multipliers to the real lifts with the partner; smaller bags if needed
S21		Biology	Toxicology				

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Air at the workstations stays below the occupational exposure limits for formaldehyde and acrolein with the hoods running			OSHA PELs: formaldehyde 0.75 ppm (8 h), acrolein 0.1 ppm; emission rates of small open machines are not published	Hoods sized for 0.5 m/s; exposure can only be shown by air monitoring on a running site	Plausible, to confirm at TRL 3	Open: air monitoring on a running site (TRL 4, on hold)
S22	Washed flake on 7.2 m2 of trays dries by the next shift with the 0.4 kW fan running through the 8 h shift	Physics	Thermodynamics	Psychrometric moisture pick-up: about 0.3 m3/s past the trays at 4 g of water per kg of air, against 30 % water on the day's flake over the 8 h the fan runs (still air overnight not counted)	1.97 kg/h <= 5.2 kg/h	Within known science	RFE-CAL-001 section 3
S23	The shredder's assumed 15 kg/h lies inside the published 8.8 to 41.8 kg/h range for its class	Engineering and technology	Standards practice	Precious Plastic Shredder Pro published range	15 kg/h <= 41.8 kg/h	Within known science	RFE-CAL-001 section 3 (rate measured at TRL 4)
S24	The extruder's assumed 5 kg/h lies at the low end of the published 5 to 15 kg/h for the Extrusion Pro class	Engineering and technology	Standards practice	Precious Plastic Bazar listing, V4 extrusion: 5 to 15 kg/h at 4 kW, 400 V	5 kg/h >= 5 kg/h	Within known science	RFE-CAL-001 section 3
S25	Every passport record validates against schema v0.2, and a product lot without	Engineering and technology	Standards practice	JSON Schema draft 2020-12 validation of the example records (RFE-CAL-001 section 11)	3 of 3 examples valid; the negative case is rejected	Within known science	RFE-CAL-001 section 11

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	parents is rejected						
S26	The site breaks even at a product price of about 2.49 USD/kg (arithmetic only: the inputs, including the 2.50 USD/kg price, are assumptions)	Mathematics	Orders of magnitude	Hand recomputation: (all costs per shift including equipment recovery - other sales) / product mass	claimed 2.49 USD/kg, estimate 2.49 USD/kg (ratio 1.00)	Within known science	RFE-CAL-001 section 10
S27	A lithium cell crushed in the shredder can start a fire, so cells are pulled out at intake and kept in dry sand	Chemistry	Electrochemistry and batteries	US EPA, An Analysis of Lithium-ion Battery Fires in Waste Management and Recycling, 2021	Intake rule, fire-safe sand container in the cage, BLD stop S5	Within known science	docs/ playbook/ safety.md
B1	Option B: a 10 t vertical baler makes bales of about 120 kg of sorted PET bottles	Physics	Mechanics and statics	Bale mass = chamber volume (1.0 x 0.7 x 0.75 m) x bale density, taken as about 230 kg/m3 to match the supplier listing's 120 kg (so not an independent estimate); platen pressure only about 0.14 MPa	claimed 120 kg, estimate 121 kg (ratio 0.99)	Plausible, to confirm at TRL 3	Open: bale density on a trial bale (TRL 4, on hold)
B2	Option B: a 2.2 kW single-phase power pack drives the 10 t ram at a usable speed	Physics	Fluid mechanics	Ram speed v = P eta / F (eta 0.7) against a usable minimum of about 10 mm/s (a 1 m stroke in under 2 min)	15.7 mm/s >= 10 mm/s	Within known science	RFE-OPT-001
B3	Option B: the baler, fans and lights run on the 40 A	Physics	Electromagnetism and circuits	NEC Table 430.248: 3 hp (2.2 kW) single-phase motor 17 A at 230 V; lights 1.7 A; a 0.4 kW fan 4.9 A	23.6 A <= 40 A	Within known science	RFE-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	single-phase supply						
B4	Option B: four people sort and bale 300 kg of input in a shift	Mathematics	Orders of magnitude	Three sorters at 25 kg/h for 4 h (the rate in RFE-CAL-001) with the fourth weighing, baling and issuing passports	300 kg >= 300 kg	Within known science	RFE-OPT-001
B5	Option B: a 120 kg bale is never lifted by hand; it moves on a pallet truck	Biology	Ergonomics	ISO 11228-1 reference mass 25 kg; bales and pallets on a hand pallet truck	0 kg lifted <= 25 kg lifted	Within known science	RFE-OPT-001

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. Counts: 25 within known science, 6 plausible (S3, S15, S17, S20, S21, B1), 1 fails with a fix proposed (S14), none fails.

Fixes under the fix rule

Three rows failed when the check was first run on the design as it stood on 2026-10-02.

Row	What failed	Fix	Where it was made	Record
S13	The energy budget used the motors' rated shaft power as if it were the electricity drawn, so it left out motor and drive losses	Divide shaft energy by the motor and drive efficiencies (IE3 minimum values, a 96 % drive, a 75 % single-phase pump motor). Energy per shift 42.9 to 45.6 kWh; 0.83 to 0.88 kWh/kg, still under the 1.0 kWh/kg target; result per shift +0.57 to +0.17 USD	`docs/04-calcs/sizing.py`, RFE-CAL-001 v0.6, `docs/playbook/economics_inputs.csv`, RFE-PRS-001	RFE-DEC-001 D2, decided under Amish's standing instruction of 2026-10-23
S17	The 460 x 160 mm feed slot is taller than 120 mm, the largest opening ISO 13857 Table 4 covers; above it the standard sends the designer to the reaching-over distances, so the 850 mm check did not apply	Lower the slot to 460 x 120 mm. The 1,060 mm path from the slot to the cutters now passes the Table 4 distance of 850 mm by 210 mm; the hood is 140 mm tall instead of 180; feed pieces are crushed flat or cut below 120 mm at the sorting table	`cad/src/model.py`, RFE-DWG-109 Rev P3, RFE-BLD-001 v0.4, `bom/bom.csv` line 3, RFE-CAL-001 section 7, safety rules	RFE-DEC-001 D3, decided under Amish's standing instruction of 2026-10-23
S14	The supply check divided kilowatts by 230 V at unity power factor (8.75 kW, 38.0 A). Drives with a single-phase diode input and single-phase motors draw much more current than that	Every fix changes an input Amish decided on 2026-09-25 (single-phase supply with staggered heating) or the throughput: (a) a 400 V three-phase supply, about 18.5 A a phase; (b) keep single phase and split the morning into two blocks, so the shredder and the extruder never run together, for a working day of about 10.6 h; (c) keep 8 h and single phase and accept less input per shift	Calculation added to `docs/04-calcs/sizing.py` (section D) and RFE-CAL-001 section 5; PR-017 not met	RFE-DEC-001 D1, proposed, awaiting Amish (recommendation: a)

Fields not relevant

All five fields have rows. Within them, nuclear and ionising radiation, optics, ecology and microbiology are not relevant: nothing radioactive or optical is processed, nothing is released to land or water (wash water and sludge go to a licensed route), and the wash water's microbial load is handled by gloves and the refusal of medical waste rather than by a design figure.

Concept options

- **Option A, reference micro-factory (the existing design, proposed to stay chosen):** S1 to S27. Open on S14 (supply current, to Amish) and on the plausible rows S3, S15, S17, S20 and S21.
- **Option B, sort-and-bale hub:** B1 to B5. Within known science; it needs about 24 A on the same single-phase supply because nothing is heated and only one motor runs. The bale mass rests on an assumed bale density until a trial bale is weighed (B1).

Independent review

On 2026-10-25 a second agent, given only this document, the calculation note RFE-CAL-001, the product requirements RFE-PRS-001 and `docs/04-calcs/sizing.py`, and not the author's reasoning, tried to refute the headline results and every "Within known science" verdict. It recomputed the mass balance, energy, press heating, fan, hood, noise, current and reach figures and could not refute them; it confirmed S14 fails. Its findings and what changed:

Row	Finding	Change
S3	The 0.8 kW standing loss is an assumption checked against an estimate, for a closed press only	Verdict to Plausible; closed by measurement at TRL 4
S4	The check used average power over the cycle, not the heating phase	Restated: at full power the sheet's heat goes in within 36.6 min of the 55 min cycle (cooling is in the cooling press); still within
S10	NFPA 92 is a fire-plume equation, used far below its range; the hot mould is a bigger short source	Cited as a heuristic; mould unloading noted
S11	ACGIH does not give a 10 m/s minimum for fumes	Restated as a design choice (ACGIH about 5 to 10 m/s); wax condensation in the duct added to the design FMEA (F5a)
S14	55 A adds table currents arithmetically (conservative); block B is 91 % of the supply; option (b) costs about 13 USD a shift in wages and turns the result to about -12.7 USD; the MSEDCL source is from 2010	All added to RFE-CAL-001 section 5 and RFE-DEC-001 D1; the fail stands
S16	The open-area limit counts only the feed slot	Noted; S15 stays plausible
S17	A machine-specific type C standard (EN 12012 series) takes precedence over ISO 13857; the cutter height is assumed	Verdict to Plausible until EN 12012 is checked; added to RFE-REG-001 (G8a)
S18	Bending only; the service deflection is about 41 mm (L/290) and racking stiffness is unchecked	Both stated for the structural engineer
S20	25 kg is the ISO 11228-1 reference mass under ideal conditions, not a limit for repeated lifts at height	Verdict to Plausible
S22	The fan runs 8 h, not overnight, and 5 % water is optimistic	Rerun with the fan's 8 h and 30 % water: 1.97 kg/h needed against 5.2 kg/h; still within
S26	The check re-adds the model's own inputs	Restated as arithmetic only; the product price is the assumption that matters (RFE-MKT-001)
S2	Float-sink does not split HDPE from PP	Stated: they are sorted into their own bins at the table
B1	The bale density was chosen to match the listing, so the check is circular	Stated; stays plausible
CAL	Meeting PR-008 with cleaner loads raises the shredder load to 4.26 h (PR-006); passport validation uses invented records; results.csv lacked PR-018 to PR-020; energy leaves out hot wash, centrifuge and conveyors	All added to RFE-CAL-001 and RFE-PRS-001; results rows added

Sources

- ABB, *Technical note: IEC 60034-30-1 standard on efficiency classes for low voltage AC motors*, 2018, Table 1 (checked 2026-10-25): https://library.e.abb.com/public/db64d153e3c346938e18916e66fb1d0d/9AKK107319%20EN%2005-2018_20848_ABB_Technical_note_IEC_60034_30_1.pdf
- Schneider Electric, ATV12HU22M2 product data sheet, line current 20.2 A at 240 V (checked 2026-10-25): https://media.cetnaj.com.au/Datasheets/Schneider/Schneider_ATV12HU22M2.pdf
- Schneider Electric, ATV12HU15M2 product data sheet, line current 14.9 A at 240 V (checked 2026-10-25): https://www.mouser.com/datasheet/2/357/Altivar_12_ATV12HU15M2-1782510.pdf
- NFPA 70, *National Electrical Code*, 2020, Table 430.248, as reproduced by Elliott Electric Supply (checked 2026-10-25): https://elliottelectric.com/StaticPages/ElectricalReferences/ElectricalTables/Amperes_Single_Phase_Alternate.aspx
- MSEDCL, *Conditions of Supply*, 2010, clause 6.4.1: single-phase supply up to 40 A, three-phase above (checked 2026-10-25): https://www.mahadiscom.in/wp-content/uploads/2018/08/CoS_MSEDCL_2010.pdf
- Troax, *Safety guide* (ISO 13857 Tables 4 and 7) and Axelent, *Standards* (openings over 120 mm use the reaching-over distances) (checked 2026-10-25): https://www.troax.com/app/uploads/Brochure-Safety-Guide_EN.pdf; https://www.axelent.dk/media/3343/normsen.pdf
- TIS (GDV), *Permissible loading capacity of containers*: 40 ft floor line load 3.0 t/m (checked 2026-10-25): https://www.tis-gdv.de/tis_e/containe/belast/belast-htm/
- Precious Plastic Bazar, 1 x 1 m hot sheet press, 430 kg (checked 2026-10-25): https://bazar.preciousplastic.com/machines/sheetpress/sheet-press-fully-built/1x1-meter-hot-sheet-press
- Precious Plastic Bazar, V4 extrusion, 5 to 15 kg/h at 4 kW, 400 V (checked 2026-10-25): https://bazar.preciousplastic.com/machines/extruder-pro/extruder-pro-fully-built/v4-extrusion/
- Precious Plastic Academy, Shredder Pro and sheetpress pages (as cited in RFE-CAL-001 section 3).
- G. Szarka and B. Iván, "Thermal properties, degradation and stability of poly(vinyl chloride) predegraded thermooxidatively in the presence of dioctyl phthalate plasticizer," *Journal of Macromolecular Science, Part A* 50, no. 2 (2013): 208 to 214 (checked 2026-10-25): https://real.mtak.hu/4463/
- US EPA, *An Analysis of Lithium-ion Battery Fires in Waste Management and Recycling*, 2021 (checked 2026-10-25): https://www.epa.gov/system/files/documents/2021-08/lithium-ion-battery-report-update-7.01_508.pdf
- Borealis, *Statement on impurities: polyolefins*, ed. 6, 2022; OSHA Table Z-1; Plastics Technology, "How to Dry PET for Container Applications"; RJC Mold, "Plastic Melting Point Chart"; KiTech float-sink note (all as cited in RFE-CAL-001 and the recipes).
- NFPA 92, *Standard for Smoke Control Systems*, axisymmetric plume equation; ACGIH, *Industrial Ventilation: A Manual of Recommended Practice for Design*, minimum duct design velocities; B. Wunderlich, *Thermal Analysis of Polymeric Materials* (2005), heat of fusion of fully crystalline polyethylene, 293 J/g; IS 875 (Part 2), imposed loads on roofs; ISO 11228-1, manual lifting.
- Section properties of UB 305 x 165 x 40 (depth 303.4 mm, width 165.0 mm, plastic modulus 623 cm3): steel section tables to EN 10365.